TMLR paper: old vs revised data

Learning Long-Range Representations with Equivariant Messages

Each table cell shows either the unchanged value or ~~*old*~~ *new* when the number shifted. Old values are the originally-published numbers; new values are the revised numbers.

Biodimers totals

Per-class indexing bug in the LOREM evaluation driver. Totals over the full biodimers test set are essentially unchanged (within 3-decimal rounding) for every variant, as expected: a per-class indexing error cannot move the aggregate. The starred (†) ablation seeds were unaffected. Per-class results, Fig. 1, changed significantly.

Table 1: Biodimers test-set totals across LOREM variants.

	Biodimers (test set, total)			
Variant	E RMSE (meV/at)	E MAE (meV/at)	**F** RMSE (meV/Å)	**F** MAE (meV/Å)
LOREM	0.222	0.155	1.646	1.023
LOREM†	0.370	0.322	1.985	1.110
LOREM LR $l = 0$	0.302	0.268	1.677	1.010
LOREM LR $l = 0^\dagger$	0.431	0.255	1.838	1.124
LOREM LR $l = 1$	0.329	0.203	1.725	0.981
LOREM LR $l = 1^\dagger$	0.329	0.302	1.698	1.043
LOREM no LR	8.259	3.784	16.452	5.468
LOREM no LR†	8.978	3.934	16.381	5.588

The per-class breakdown is where the bug actually shows; Fig. 1 reproduces the main-text MAE plot with an extra faded LOREM bar carrying the originally-published per-class numbers.

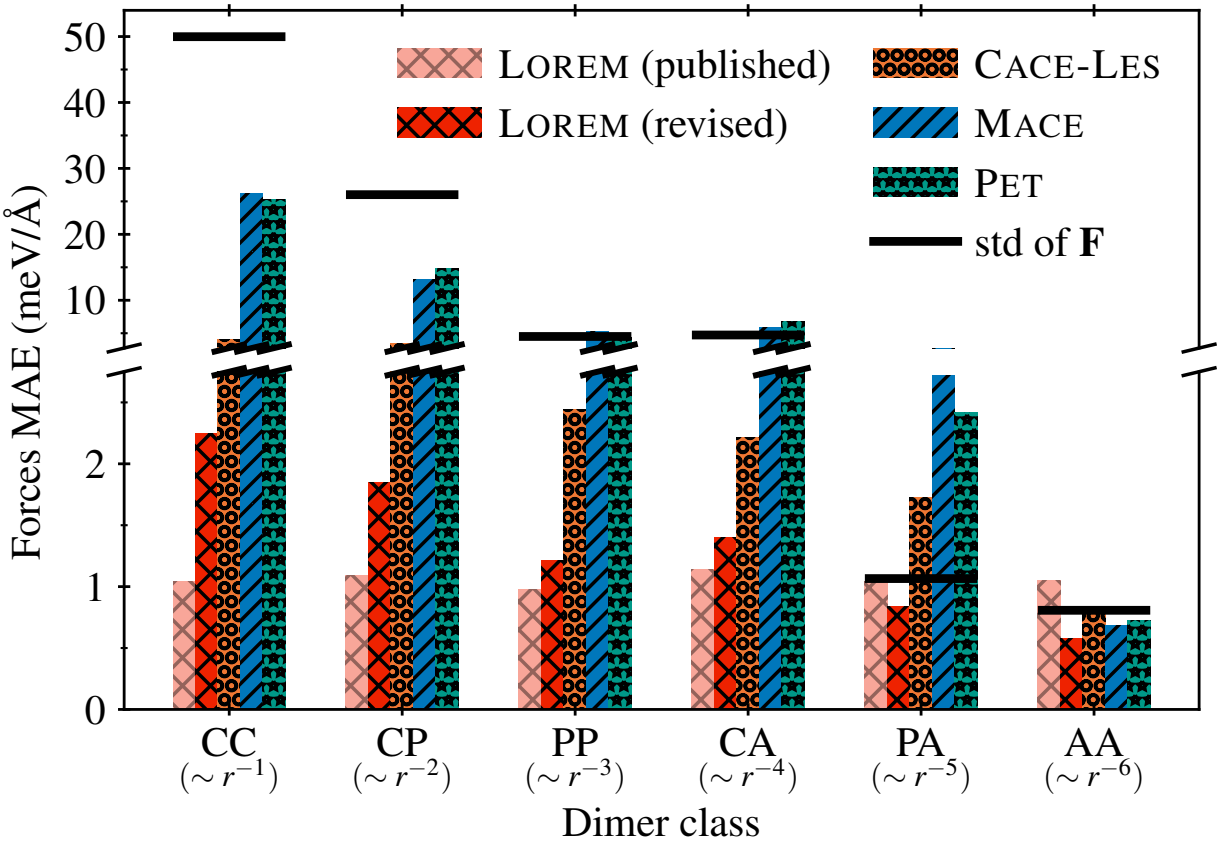


Figure 1: Per-class force MAE on biodimers (the main-text figure). The faded LOREM bar in each class group is the originally-published number; the bold LOREM bar is the corrected value. CACE-LES, MACE, and PET are unchanged. Source: `results/biodimers_figure_comparison.py`.

Benchmark timings (Appendix E)

The Ewald-vs-PME timing tables shifted because the original benchmark ran Ewald and PME at mismatched accuracy. The revised *light* regime fixes the PME accuracy target while letting Ewald reach larger N thanks to the smaller k-grid. `nolr` was also re-run so the LR-fraction breakdown is apples-to-apples.

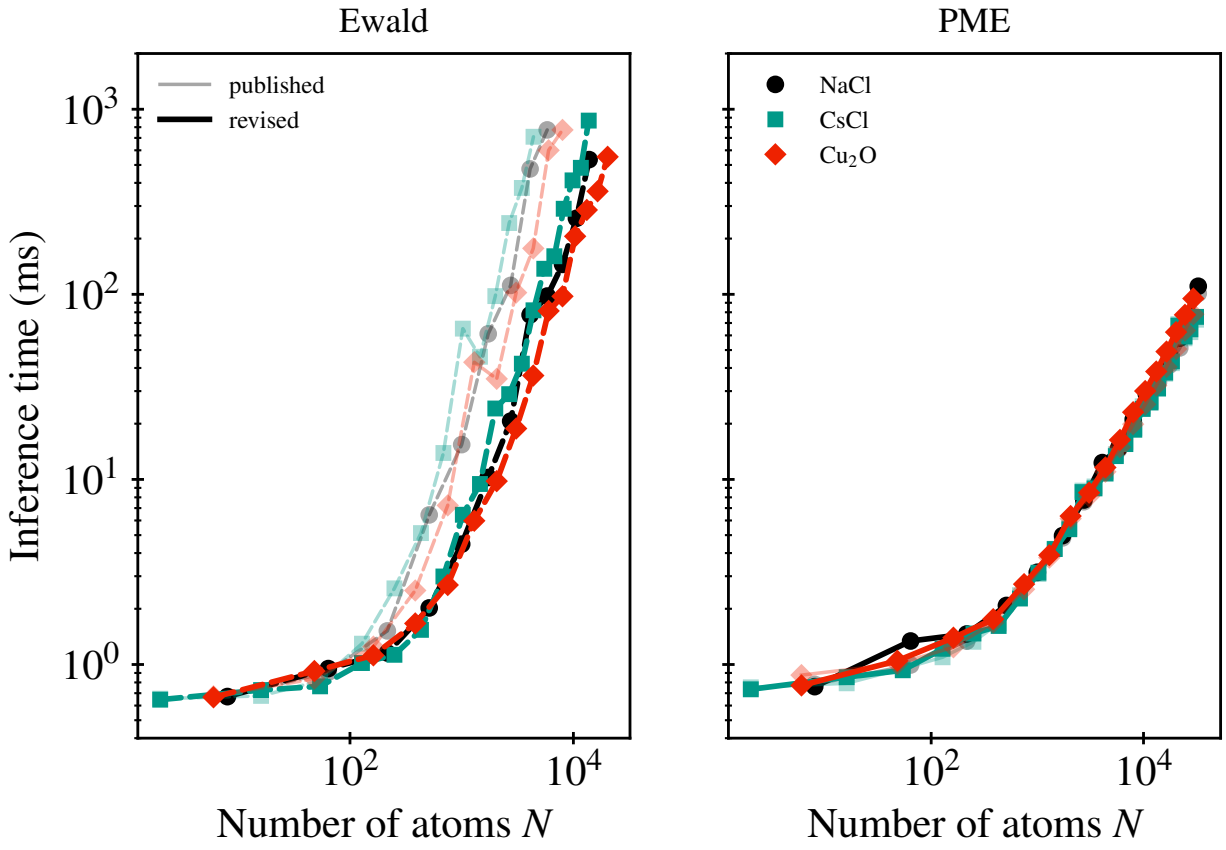


Figure 2: Light-regime benchmark: original (faded) vs revised (bold) timings for all three crystals. Ewald dashed, PME solid. Source: `results/benchmark_comparison_figure.py`.

Table 2: NaCl: Ewald vs PME, LR breakdown.

	Ewald			PME		
N	SR+LR (ms)	LR (ms)	LR (%)	SR+LR (ms)	LR (ms)	LR (%)
8	0.7	~~0.2~~ 0.3	~~32.4~~ 40.7	0.8	~~0.3~~ 0.4	~~41.9~~ 47.6
64	~~0.9~~ 1.0	0.4	~~45.4~~ 43.1	~~1.0~~ 1.3	~~0.5~~ 0.8	~~50.0~~ 59.6
512	~~6.4~~ 2.0	~~5.3~~ 0.9	~~82.6~~ 42.9	2.1	~~1.0~~ 0.9	~~46.0~~ 44.6
1728	~~61.2~~ 10.2	~~58.4~~ 7.3	~~95.5~~ 71.3	~~4.8~~ 5.0	~~2.1~~ 2.0	~~42.6~~ 40.8
4096	~~475.8~~ 77.5	~~469.8~~ 71.0	~~98.7~~ 91.6	~~12.1~~ 12.3	~~6.1~~ 5.8	~~50.5~~ 47.2
8000	– 145.1	– 132.3	– 91.2	~~20.2~~ 21.1	~~8.9~~ 8.3	~~43.8~~ 39.3
17576	–	–	–	~~41.5~~ 46.3	~~16.9~~ 15.7	~~40.7~~ 33.9
32768	–	–	–	~~101.2~~ 110.6	~~51.6~~ 46.9	~~51.0~~ 42.4

Table 3: CsCl: Ewald vs PME, LR breakdown.

	Ewald			PME		
N	SR+LR (ms)	LR (ms)	LR (%)	SR+LR (ms)	LR (ms)	LR (%)
2	~~0.7~~ 0.6	0.3	~~42.7~~ 40.4	~~0.8~~ 0.7	~~0.4~~ 0.3	~~50.2~~ 47.6
54	0.8	~~0.4~~ 0.3	~~48.1~~ 39.9	0.9	0.5	~~55.4~~ 50.8
432	~~5.1~~ 1.5	~~4.3~~ 0.7	~~84.5~~ 48.1	~~1.7~~ 1.6	~~0.9~~ 0.8	~~52.4~~ 50.5
1458	~~46.0~~ 9.4	~~44.3~~ 7.6	~~96.2~~ 80.5	4.2	~~2.5~~ 2.4	~~58.8~~ 56.2
4394	~~710.1~~ 82.0	~~705.7~~ 77.4	~~99.4~~ 94.4	~~11.0~~ 10.8	~~6.7~~ 6.1	~~60.6~~ 57.1
8192	– 289.9	– 281.6	– 97.1	~~19.0~~ 18.6	~~11.2~~ 10.2	~~59.2~~ 55.2
18522	–	–	–	~~42.5~~ 43.4	~~25.5~~ 23.5	~~60.0~~ 54.0
31250	–	–	–	~~72.9~~ 75.5	~~44.2~~ 41.0	~~60.7~~ 54.3

Table 4: Cu_2O: Ewald vs PME, LR breakdown.

	Ewald			PME		
N	SR+LR (ms)	LR (ms)	LR (%)	SR+LR (ms)	LR (ms)	LR (%)
6	0.7	0.3	~~42.2~~ 43.4	~~0.9~~ 0.8	~~0.5~~ 0.4	~~55.6~~ 51.1
48	~~0.8~~ 0.9	~~0.4~~ 0.5	~~42.8~~ 49.3	1.0	~~0.5~~ 0.6	~~52.5~~ 55.3
384	~~2.5~~ 1.7	~~1.5~~ 0.5	~~58.6~~ 32.7	~~1.7~~ 1.8	0.6	~~38.4~~ 36.2
1296	~~43.0~~ 6.0	~~40.3~~ 3.1	~~93.9~~ 52.3	~~3.8~~ 3.9	~~1.1~~ 1.0	~~30.4~~ 26.8
4374	~~177.1~~ 36.4	~~169.1~~ 27.7	~~95.4~~ 76.0	~~11.0~~ 11.6	2.9	~~26.5~~ 24.6
7986	~~773.1~~ 97.6	~~759.0~~ 80.0	~~98.2~~ 82.1	~~19.9~~ 23.1	~~5.7~~ 5.6	~~28.7~~ 24.1
16464	– 360.4	– 322.8	– 89.6	~~40.5~~ 49.1	~~11.3~~ 11.4	~~27.8~~ 23.3
29478	–	–	–	~~78.1~~ 94.8	~~19.6~~ 18.9	~~25.1~~ 19.9